Opgave: Presningspasning (MEM1)

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$p_c := 40\ MPa$ $R_f := 50\ mm$ $r_o := 80\ mm$ $E := 200\ GPa$

## Del 1 - Formlen for samme materiale og massiv aksel

Fladens radius hedder $R_f$ (R er gaskonstanten i Mathcad). Akslen er massiv ($r_i = 0$), og begge dele er af stål, så Poissons forhold går ud. Slides forenkler det radiale overlap til (Kapitel10_Slides, slide 18):

$$\delta = \frac{p \cdot R}{E}\left(\frac{r_o^2 + R^2}{r_o^2 - R^2} + 1\right)$$

## Del 2 - Overlappet

$\frac{p_c \cdot R_f}{E} = 0.01\ mm$ $\frac{\left(r_o^{\ 2} + R_f^{\ 2}\right)}{\left(r_o^{\ 2} - R_f^{\ 2}\right)} = 2.282$

$\delta_r := \frac{p_c \cdot R_f}{E} \cdot \left(\frac{\left(r_o^{\ 2} + R_f^{\ 2}\right)}{\left(r_o^{\ 2} - R_f^{\ 2}\right)} + 1\right) = 32.821\ \mu m$

Det er det radiale overlap. På diameteren er det det dobbelte:

$2 \cdot \delta_r = 65.641\ \mu m$

$\delta_r = 32.821\ \mu m$

### Worked Example — Sizing an Interference Fit Joint

Evaluate required radial interference ($\delta$) to establish contact pressure $p = 40$ MPa at interface ($R = 50$ mm) of a solid steel shaft inside a thick steel hub ($r_o = 80$ mm).

1. Given Mechanical Design Parameters

Since shaft is solid ($r_i = 0$) and both parts are steel ($E = 200$ GPa $= 200\,000$ MPa):

$R = 50$ mm, $r_o = 80$ mm, $r_i = 0$, $p = 40$ MPa

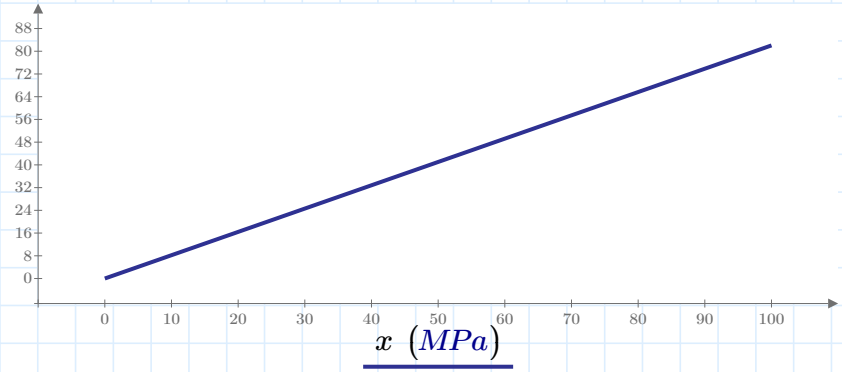
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**Plot og tjek dig selv - Dobbelt tryk**

δ er proportional med p, så dobbelt kontakttryk kræver dobbelt overlap:

$$\delta_p(x) := \frac{x \cdot R_f}{E} \cdot \left( \frac{\left(r_o^{\,2} + R_f^{\,2}\right)}{\left(r_o^{\,2} - R_f^{\,2}\right)} + 1 \right) \qquad x := 0\ MPa, 1\ MPa .. 100\ MPa \qquad \delta_p(80\ MPa) = 65.641\ \mu m$$

En ret linje gennem origo: 40 MPa kræver 32,8 µm, og 80 MPa kræver 65,6 µm.




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